温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Simple Linear Regression I

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Review of Hypothesis Testing

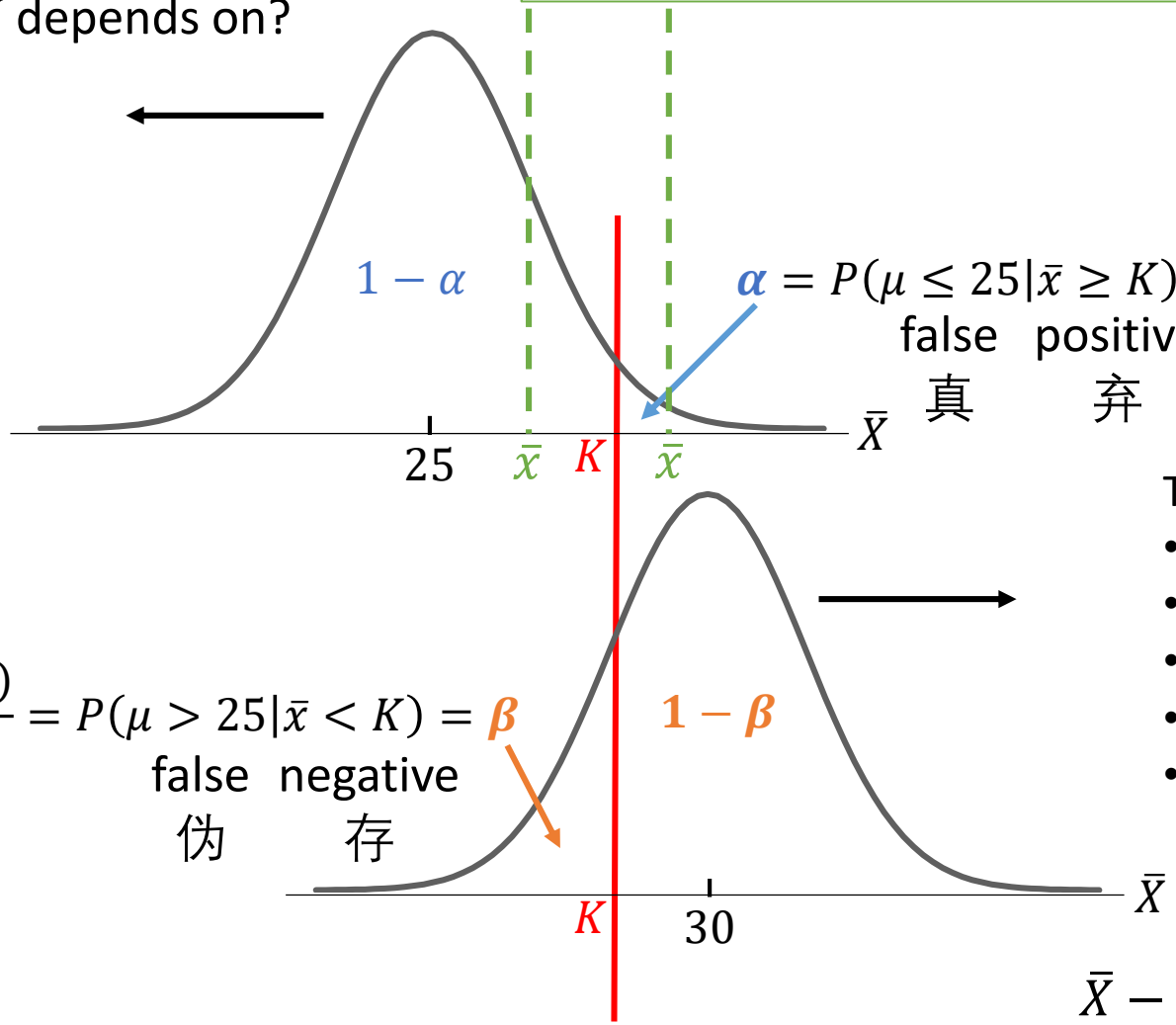
What is p-value = $P(\bar{X} \geq \bar{x} | \mu \leq 25)$?
Measure the **fit** between **sample** data and $\boldsymbol{H_0}$.

Decision Rule: Reject $\boldsymbol{H_0}$ if $\bar{\boldsymbol{x}} \geq \boldsymbol{K}$.

<u>Think:</u> **Critical value $\boldsymbol{K}$** depends on?

$$H_0\colon\ \mu \leq 25$$

$$H_1\colon\ \mu > 25$$



$$\boldsymbol{\alpha} = P(\mu \leq 25 | \bar{x} \geq K) = \frac{P(\bar{x} \geq K | \mu \leq 25) P(\mu \leq 25)}{P(\bar{x} \geq K)}$$

false positive
真 弃

$$\frac{P(\bar{x} < K | \mu > 25) P(\mu > 25)}{P(\bar{x} < K)} = P(\mu > 25 | \bar{x} < K) = \boldsymbol{\beta}$$

false negative
伪 存

Think:

- Which one is more important: α, β?
- $\alpha \uparrow,\ \beta \downarrow$
- $\alpha + \beta = 1$?
- β depends on population mean.
- How sample size $\boldsymbol{n}$, population std $\boldsymbol{\sigma}$ influence $\boldsymbol{K}$?

$$\bar{X} \sim N(\mu, \sigma/\sqrt{n}) \qquad \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1} \qquad t \approx z \text{ if } n > 30$$

Motivating Example

Do more study hours lead to higher GPA?

- **Scenario**

  At WKU, we collected data from a large group of students on how much time they study each week and their overall GPA. From this population, we randomly selected **100** to analyze.

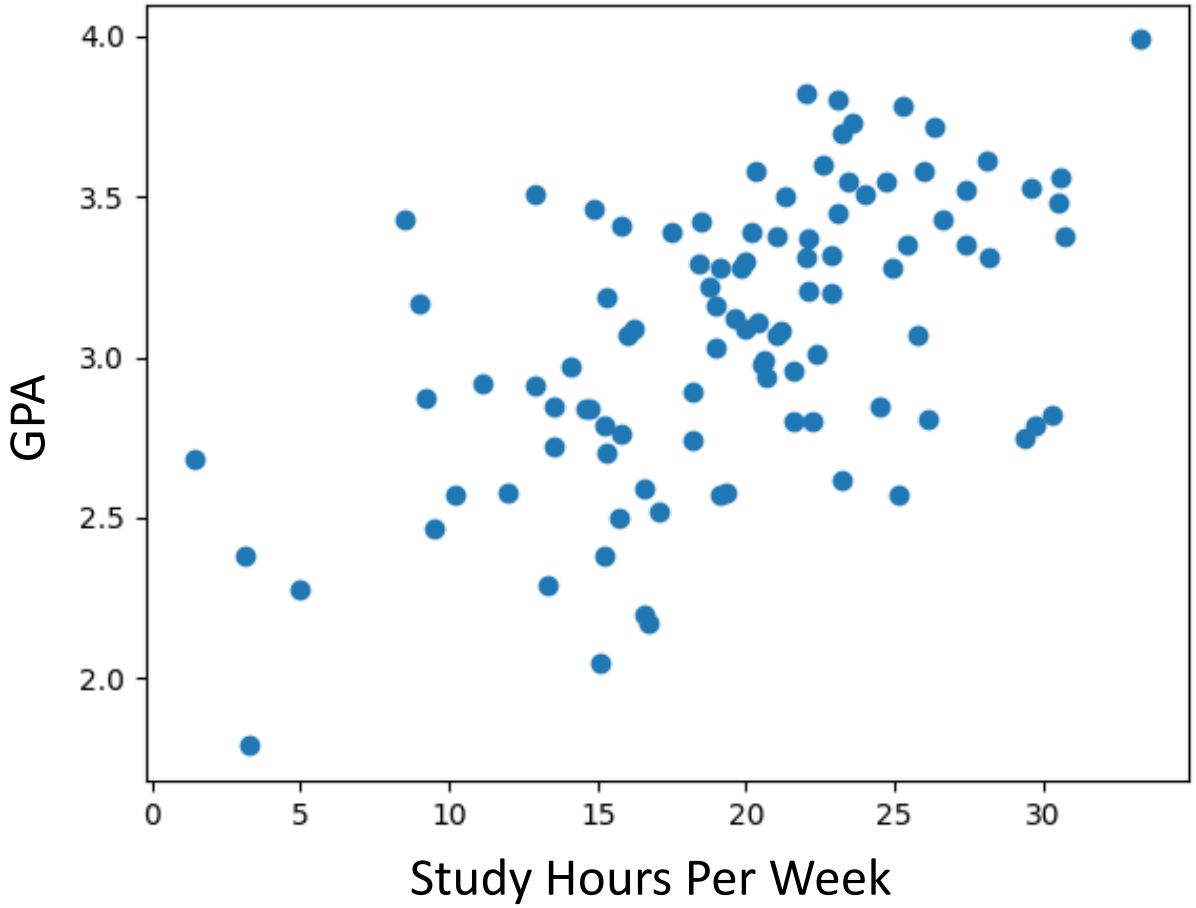
- Each observation represents one student
  - **Hours (X):** how many hours a student studies on average per week
  - **GPA (Y):** the student's overall grade point average (0 – 4.0 scale)

Hours	GPA
16.0	3.07
9.0	3.17
22.0	3.31
24.9	3.28
21.0	3.07

	Hours	GPA
mean	19.65	3.07
std	6.42	0.44
min	1.40	1.79
50%	20.25	3.09
max	33.30	3.99

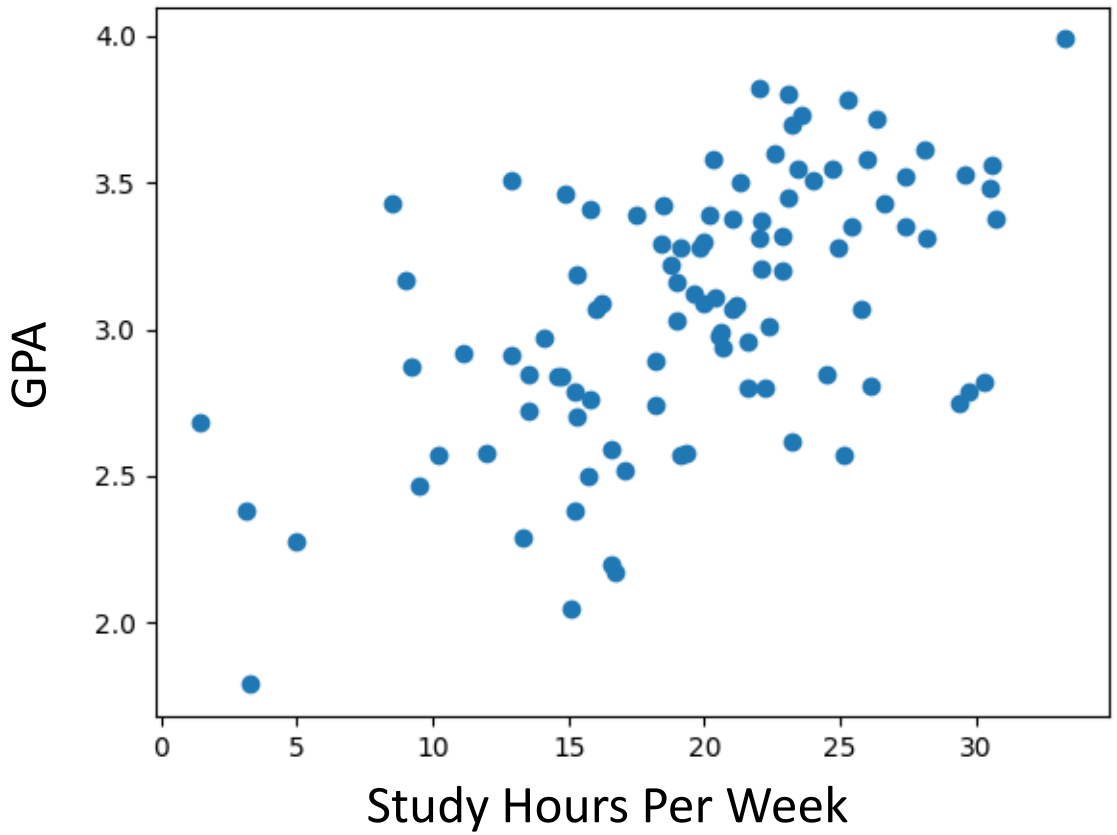
Motivating Example

What insights can you draw from this scatter plot?



Motivating Example

Do more study hours lead to higher GPA?



- Is there a real, statistically significant **relationship** between study hours and GPA?
- How much does GPA increase, on average, when students study one more hour per week?
- Can we predict a student's GPA based on their study time?

What Is Regression Analysis

Regression analysis is a statistical procedure that uses data to develop an equation showing how the variables are related.

- Dependent (or response) variable (**DV**) is the variable being predicted.
- Independent (or predictor) variables (or features) (**IV**) are variables used to predict the value of the dependent variable.
- Simple linear regression is a form of regression analysis in which a ("simple") single independent variable, x is used to develop a "**linear**" relationship (straight line) with the dependent variable, y.

$$\boldsymbol{y = \beta_0 + \beta_1 x + \varepsilon}$$

- Multiple linear regression is a more general form of regression analysis involving two or more independent variables.

$$\boldsymbol{y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \varepsilon}$$

Simple Linear Regression Model

- The simple linear regression model is an equation that describes how the dependent variable y is related to the independent variable x and error term ε.

$$\boldsymbol{y = \beta_0 + \beta_1 x + \varepsilon}$$

Where

- y is the dependent variable.
- x is the independent variable.
- β_0 and β_1 are referred to as the population parameters.
- ε is the **error term**. It accounts for the variability in y that cannot be explained by the linear relationship between x and y.

Estimated Simple Linear Regression Equation

The **estimated** simple linear regression equation is described as follows:

$$\hat{\boldsymbol{y}} = \hat{\boldsymbol{\beta}}_{\mathbf{0}} + \hat{\boldsymbol{\beta}}_{\mathbf{1}}\boldsymbol{x}$$

- Where $\hat{y}$ is the **point estimator** of $E(y|x)$, the mean of y for a given x.
- $\hat{\beta}_0$ is the point estimator of β_0 and the y-intercept of the regression.
  - The y-intercept $\hat{\beta}_0$ is the estimated value of the dependent variable y when the independent variable x is equal to 0.

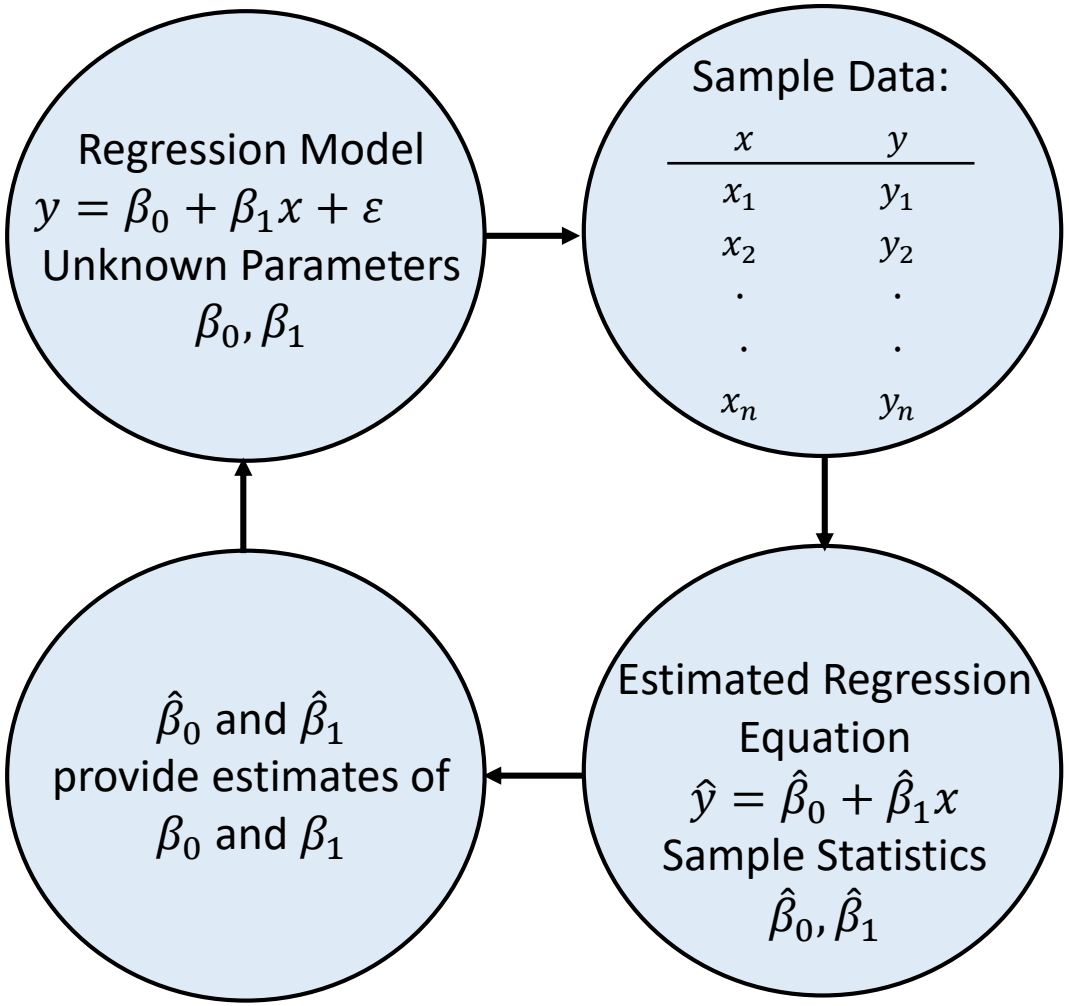
Regression coefficient

- $\hat{\beta}_1$ is the point estimator of β_1 and the slope of the regression.
  - The slope $\hat{\beta}_1$ is the estimated change in the value of the dependent variable y that is associated with a **one unit** increase in the independent variable x.

The Estimation Process in Simple Linear Regression

- The estimation of $\hat{\beta}_0$ and $\hat{\beta}_1$ is a statistical process much like the estimation of the population mean μ.

- β_0 and β_1 are the unknown parameters of interest, and $\hat{\beta}_0$ and $\hat{\beta}_1$ are the sample statistics used to estimate the parameters.

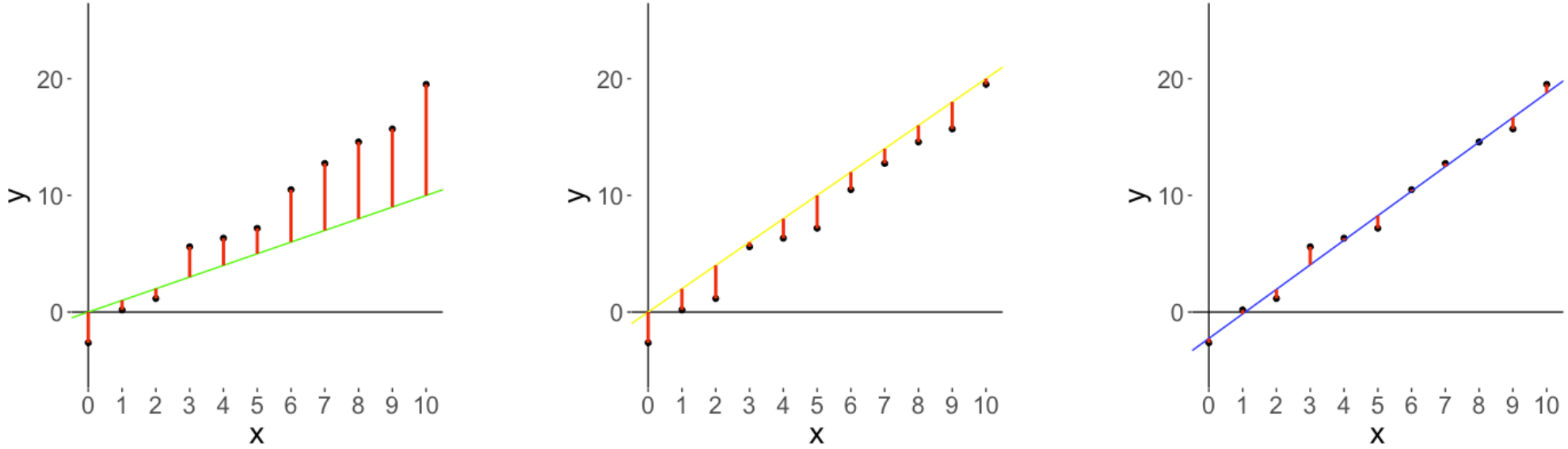


How to Compute $\hat{\beta}_0, \hat{\beta}_1$ – Ordinary Least Squares (OLS)

"True" model: $y = \beta_0 + \beta_1 x + \varepsilon$

Our model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

Define Residual / Residual Error: $e_i = y_i - \hat{y}_i$



How to Compute $\hat{\beta}_0, \hat{\beta}_1$ – Ordinary Least Squares (OLS)

Goal: Choose $\hat{\beta}_0, \hat{\beta}_1$ so as to **minimize the sum of squares of the residuals**.

$$\sum_i^n e_i^2 = \sum_i^n (y_i - \hat{y}_i)^2 = \sum_i^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

Derivation (for your edification, **not tested**): Minimize by setting derivatives to 0.

Define means: $\bar{y} \equiv \frac{1}{n}\sum_i^n y_i$, $\bar{x} \equiv \frac{1}{n}\sum_i^n x_i$

$$Z \equiv \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\frac{\partial Z}{\partial \hat{\beta}_0} = \sum_{i=1}^n -2(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\frac{\partial Z}{\partial \hat{\beta}_1} = \sum_{i=1}^n -2x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\hat{\beta}_0 = \frac{1}{n}\sum_{i=1}^n y_i - \frac{1}{n}\sum_{i=1}^n \hat{\beta}_1 x_i \Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\sum_{i=1}^n x_i y_i - \sum_{i=1}^n (\bar{y} - \hat{\beta}_1 \bar{x}) x_i - \sum_{i=1}^n \hat{\beta}_1 x_i^2 = 0$$

$$\sum_{i=1}^n x_i y_i - \bar{y}\sum_{i=1}^n x_i + \hat{\beta}_1 \left(\bar{x} \sum_{i=1}^n x_i - \sum_{i=1}^n x_i^2 \right) = 0$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - \bar{y}\sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2 - \bar{x}\sum_{i=1}^n x_i}$$

$$= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$$

$$= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

How to Compute $\hat{\beta}_0, \hat{\beta}_1$ – Ordinary Least Squares (OLS)

Goal: Choose $\hat{\beta}_0, \hat{\beta}_1$ so as to **minimize the sum of squares of the residuals**

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

OLS Solution:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{Cov(x, y)}{Var(x)}$$

OLS Estimator

No need to memorize.

Interpreting a Regression Model

Motivating Example Revisit

How do we interpret the regression coefficients?

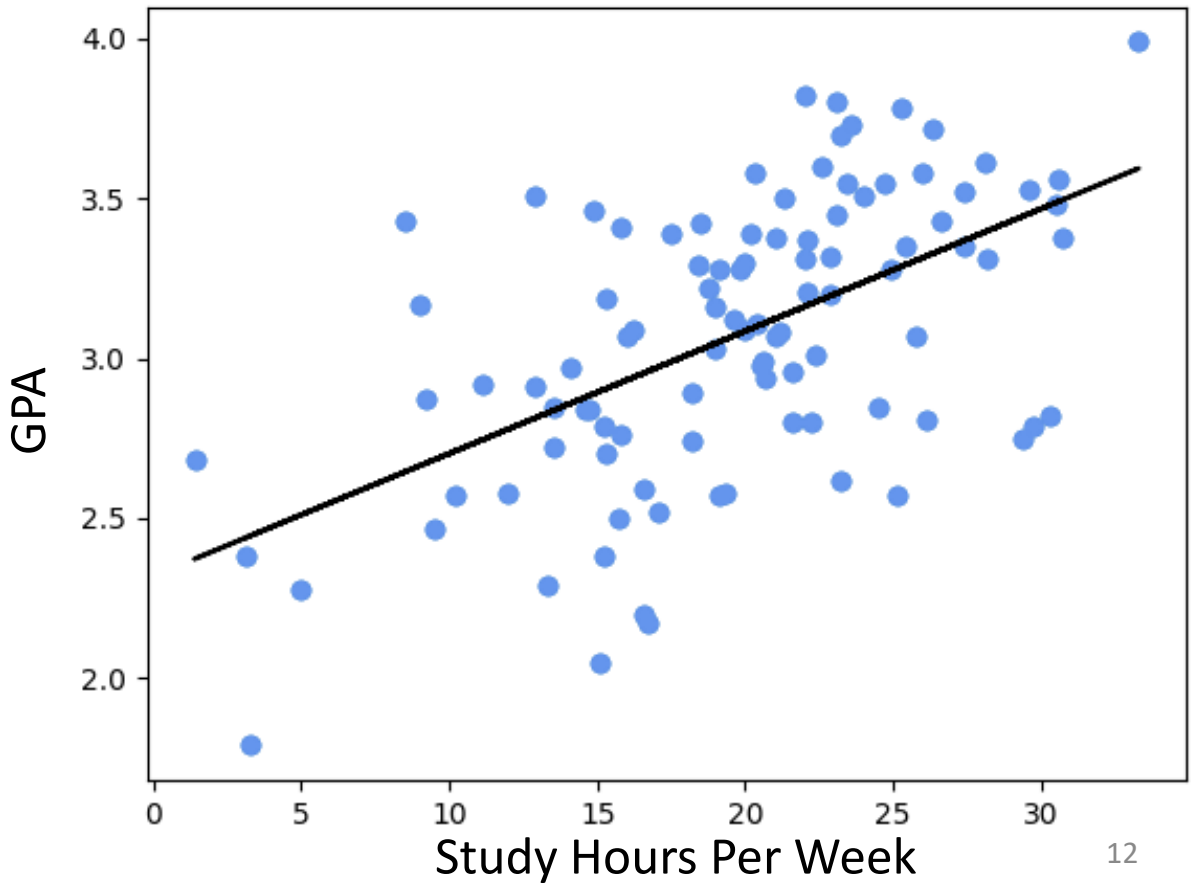
$\hat{\beta}_0$: the mean value of y when x is zero

$\hat{\beta}_1$: According to the model, a one-unit change in x results in a $\hat{\beta}_1$-unit change in y.

$$\text{GPA} = 2.32 + 0.04 * Hours$$

$\hat{\beta}_0$: The predicted GPA is 2.32 for a student who does not study at all.
$\hat{\beta}_1$: For each additional 1 hour of study per week, the GPA increases by 0.04 points on average.

$$\hat{y} = 2.32 + 0.04x$$



Experimental Region and Extrapolation

- The regression model is valid only within the range of study hours observed in our sample (the experimental region).
- **Extrapolation** — predicting GPA for study hours *outside* this range — is risky and should be avoided unless supported by additional evidence.
- In our dataset, study hours range roughly from 0 to 40 hours per week.
- Predicting GPA for, say, 60 hours of study per week is not reliable, because the model has never seen such data.
- Similarly, the intercept ($\hat{\beta}_0$) — the predicted GPA when study hours = 0 — may not have meaningful interpretation in reality.

Assessing Model Fit – Sums of Squares

- The value of the sum of squares due to error (SSE) is a measure of the error that results from using the $\hat{y}_i$ values to predict the y_i values.

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- The value of the total sum of squares (SST) is a measure of the error that results from using the sample mean $\bar{y}$ to predict the y_i values.

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- The value of the sum of squares due to regression (SSR) is a measure of how much the $\hat{y}_i$ values deviate from the sample mean $\bar{y}$.

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- The relationship between these three sums of squares is $SST = SSR + SSE$.

Assessing Model Fit – Coefficient of Determination

- The ratio SSR/SST is called the coefficient of determination, denoted by R^2.

$$\boldsymbol{R^2 = \frac{SSR}{SST}}$$

- The coefficient of determination can only assume values between 0 and 1 and is used to evaluate **the goodness of fit for the estimated regression equation**.
- R^2 measures how much of the variation in DV is explained by IV.
- A perfect fit exists when y_i is identical to $\hat{y}_i$ for every observation i so that all residuals $y_i - \hat{y}_i = 0$.
  - In such case, $SSE = 0$, $SSR = SST$, and $R^2 = SSR/SST = 1$.
- Poorer fits between y_i and $\hat{y}_i$ result in larger values of SSE and lower R^2 values.
  - The poorest fit happens when $SSE = SST$, $SSR = 0$, and $R^2 = 0$.

Summary: Simple Linear Regression

- Goal
  - Describe and predict the relationship between two variables.
- Model form

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \qquad y = \beta_0 + \beta_1 x + \varepsilon$$

- OLS Estimator
  - Ordinary Least Squares (OLS) finds the line that minimizes the sum of squared errors between observed and predicted values.

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \qquad \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{Cov(x, y)}{Var(x)}$$

- Coefficient of Determination (R^2)
  - Measures how much of the variation in DV is explained by IV.
  - Ranges from 0 to 1 — higher values mean a better model fit.